Do Channel Scores Matter? Pruning ResNet-18 for Few-Shot Aerial, Sentinel-2 and Crop Recognition

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Abstract

Small networks are what satellites and drones can carry, and the classes they meet next arrive with a few labelled examples. We re-evaluate, under one controlled protocol, a channel score (DACIS) and a prune, meta-train, prune schedule (PMP) from an earlier preprint, against seven other ways to choose which channels of a pretrained ResNet-18 to keep. Test classes are held out: 15 classes of NWPU-RESISC45 aerial scenes, five classes of 13-band Sentinel-2 EuroSAT-MS and, as a ground-level control, PlantVillage and PlantDoc leaf photographs; five class partitions, one set of test episodes for every method. Under PMP at 30% of the parameters, the eight scores are 2.6 to 3.0 points apart on four test settings. DACIS cannot be told from random selection, and only the Taylor term is ahead of it (by up to 1.9 points, on two settings), a difference that does not survive correction for the 28 comparisons. Episodic training did not raise the accuracy of the dense network. Pruning to 30% removed 58% of the multiply-accumulates but only 39% of CPU latency, while an unpruned MobileNetV3-Small, trained with the same recipe, is +7.7 and +15.3 points more accurate than the pruned ResNet-18 on RESISC45 and EuroSAT-MS at 0.018 G multiply-accumulates. For these benchmarks and this recipe, the choice of architecture and budget mattered far more than the choice of channels. A notebook reruns every number.

1. Introduction

Five labelled aerial scenes of a class the network has never seen. A ResNet-18 trained on 25 other NWPU-RESISC45 classes labels the query scenes of the new class correctly 86.9% of the time. Keep 30% of its parameters and the figure falls to 80.5% with the DACIS score, 81.0% with randomly chosen channels and 82.2% with a first-order Taylor score. This paper asks whether such differences between channel scores are real, and what else moves the number.

The question matters where a classifier has to run on small hardware that cannot be retrained often: the payload of a small satellite, a drone over a field. Onboard inference already flies [14, 15], the classes that matter change with the mission, and labels for a new class are few. A pruned network is cheap to run and a prototype head can be updated from five labelled patches without touching the weights. We measure cost only on a server CPU and GPU, so this motivation is context, not a result. The pruning studies we build on [17, 21, 23, 27] report accuracy on the classes the network was trained for. What survives when the classes at test time are not those the channels were chosen for?

We study that on ResNet-18. Every method removes the inner convolution channels of each residual block under one shared depth profile (Sec. 3), so two scores differ only in which channels they keep. The scores are random selection, filter ℓ_1 norm, BatchNorm scale, a Taylor term, and DACIS, a rank sum of gradient norm, activation variance and Fisher ratio over class labels, which we also run in its three parts. DACIS and the PMP schedule of prune, meta-train, prune again come from an earlier preprint [1]; this paper re-tests them with held-out classes, five class partitions and shared test episodes, and they do not hold up. We test on held-out classes of aerial scenes (NWPU-RESISC45 [7]) and of Sentinel-2 patches with all thirteen bands (EuroSAT-MS [18]). PlantVillage leaf photographs [20], with PlantDoc field photographs [30] as a shift check, serve as a ground-level control because the earlier preprint used them; they lack geospatial context and fall outside the Earth-observation scope of the workshop.

The contributions are four findings and the protocol that supports them.

- Under PMP at 30% of the parameters, 2 of 28 paired comparisons of a computed score with random selection favour the computed score and 3 favour random selection. Only the Taylor term is ahead, by at most 1.9 points, and no comparison survives a Bonferroni correction. DACIS cannot be told from random selection.
- Episodic training does not raise the accuracy of the dense network, and the small gain of two-stage schedules is matched by a variant that updates no weight in the meta stage.
- Pruning saves less time than operations: 58% fewer multiply-accumulates gave 39% lower single-thread CPU latency. The budget moves accuracy several times more than the score.
- An unpruned MobileNetV3-Small, trained with the same recipe, beats the pruned and the dense ResNet-18 on both Earth-observation sets at a fraction of the cost. This is a strong baseline that pruning studies in this setting should include.

The protocol (class-disjoint partitions, shared test episodes, paired intervals over partitions, the count of significant comparisons set against what chance alone gives) is meant to be reused. With five partitions it cannot exclude differences of about 1.7 points, and Sec. 6 lists what we did not test.

2. Related work

Channel pruning. Most pipelines start from one of three scores: the ℓ_1 norm of a filter [21], the BatchNorm scale learned under a sparsity penalty [23], or a first-order Taylor estimate of the loss change [27, 28]. He et al. [17] choose channels that best reconstruct the next layer's output and, for residual networks, prune only the inner convolutions of each block, which is the scheme we use. Discrimination-aware pruning [33] adds auxiliary classification losses to intermediate layers and keeps the channels that preserve them. DACIS asks a similar question with a cheaper statistic, the Fisher ratio of the pooled activation, and no extra training. These methods are evaluated on the classes they were trained for. Two earlier results make that evaluation look weaker than it is. Mittal et al. [26] showed that removing 25 to 50% of the filters of VGG-16 and ResNet-50 at random and then fine-tuning matches ranked pruning, so that recovery, not the ranking, carries the result. Liu et al. [25] found that a fine-tuned pruned network is no better than the same architecture trained from scratch. We ask the same question when the classes at test time differ from those used to choose the channels. MetaPruning [24] also uses the word "meta", but there a hypernetwork predicts weights for candidate architectures; our meta stage is ordinary episodic training whose gradients feed a later pruning step.

Is the ranking needed? Random or trivial baselines have been strong in several pruning settings. Besides the two studies above, the lottery-ticket work [13] and the survey of Blalock et al. [4], which documents weak baselines and inconsistent comparisons across pruning papers, point the same way. We found no study that compares channel scores on classes held out from the ones used to choose the channels; that gap, with paired test episodes over several class partitions, is what this paper addresses.

Few-shot classification. Prototypical networks [31] label a query by the nearest class mean in embedding space, and MAML [11] learns an initialisation that adapts in a few gradient steps. Two findings shape our protocol. A network pre-trained with plain cross-entropy on the base classes is a strong baseline [5, 32], and episodic training on top of it with a cosine nearest-centroid head can add a little more [6]. Our stages S0 and M follow that recipe. The cross-domain benchmark of Guo et al. [16] has crop-disease and EuroSAT targets and reports that meta-learning methods often lose to simple fine-tuning when the target is far from the source. Strong pretrained backbones with a simple head [9, 19] are the stronger reference point for any compact few-shot model. Section 5 asks whether that holds when the episodic stage is added to a pruned network.

Earth observation. Few-shot scene classification in remote sensing has been approached with metric learning on aerial benchmarks [22] and with meta-learning for land cover from Sentinel-2 time series [29]. NWPU-RESISC45 [7] and the Sentinel-2 [10] patches of EuroSAT [18] are standard scene and land-use benchmarks, which we split by class instead of by image. Pretraining on satellite data [8] is the natural alternative to an ImageNet initialisation; we did not test it. On-board inference is not hypothetical: CloudScout [15] runs a compact CNN on a Myriad 2 accelerator in the Φ-Sat-1 mission [14]. We do not target that hardware. Our question is narrower: when a ResNet-18 must shrink and must still recognise classes it was not trained on, does the choice of channels matter?

Plants. PlantVillage [20] photographs single leaves against a plain background. PlantDoc [30] was assembled from internet photographs because large non-laboratory plant datasets were missing, and Barbedo [3] lists image background and capture conditions among the factors that limit deep-learning disease

recognition. Argüeso et al. [2] study few-shot recognition on PlantVillage with a 32/6 class split and a Siamese network; we use a similar cross-class split with prototype classification. The method we test was first proposed for this setting, and crop monitoring is an Earth-observation task in its own right, but leaf photographs carry no geospatial context, so they are secondary here.

Earlier version of this method. A preprint [1] introduced the names PMP and DACIS and reported results from an earlier implementation. This paper re-implements both from scratch, changes the protocol (class-disjoint splits, shared depth profile, paired episodes, five partitions), and reuses none of its numbers.

3. Method

Episodes and classifier. An episode draws N=5 classes, K labelled support images and Q=15 query images per class. We embed an image with the 512-d global-average-pooled output of a ResNet-18, L2-normalise it, average the support embeddings of each class into a prototype, and give each query the label of the most similar prototype under cosine similarity. Base classes are the only classes the network sees, whether for training, for pruning decisions or for the calibration statistics below. Novel classes appear at evaluation and nowhere else.

What is pruned. A ResNet-18 has eight basic blocks, each with two 3×3 convolutions. We remove output channels of the first convolution, together with their BatchNorm entries and the matching input slices of the second (1920 "mid" channels in total). The stem, the shortcuts and the block outputs keep their width, so no index has to be matched across a residual addition; He et al. [17] prune residual branches the same way. The budget is the share of the 11.18 M trunk parameters that remain, counted on the pruned network (the classifier is a parameter-free prototype head and is not counted). To spread a budget over depth, every score in this paper uses one rule: block $b \in \{0, \dots, 7\}$ keeps a fraction $\kappa\,(1 - \tau\, b/7)$ of its mid channels, with τ=0.5 and κ found by bisection. Deep blocks hold most of the weights, so they lose most of the channels. Since the rule is shared, two scores differ only in which channels they keep inside a block.

Scores. All scores are computed on a calibration set of 40 images per base class, without augmentation. Four are standard: random; the ℓ_1 norm of the filter [21]; the BatchNorm scale $|\gamma|$ [23]; and a first-order Taylor term [27], the mean over images of $|\sum_{x,y} a\, \partial L/\partial a|$ for the post-ReLU mid activation a. The loss L is the cross-entropy of a cosine nearest-prototype classifier whose prototypes are the calibration class means under the current network, so no trained head has to survive the earlier stages.

DACIS combines three per-channel quantities. $\mathcal{G}_c$ is the mean ℓ_2 norm of $\partial L/\partial W_c$ over calibration batches of 64 images, where W_c is the filter of channel c. $\mathcal{V}_c$ is the variance, across calibration images, of the spatially pooled activation of the channel. $\mathcal{D}_c$ is the Fisher ratio of that pooled activation over the class labels, $\mathcal{D}_c = \sum_k n_k(\mu_{kc} - \mu_c)^2 / \sum_k \sum_{i \in k}(a_{ic} - \mu_{kc})^2$, the between-class over within-class scatter used in linear discriminant analysis [12]. Each quantity is replaced by its rank inside the block, scaled to $[0, 1]$ (written rk), and the score is

$$\mathrm{DACIS}_c = \lambda_G\, \mathrm{rk}(\mathcal{G}_c) + \lambda_V\, \mathrm{rk}(\mathcal{V}_c) + \lambda_D\, \mathrm{rk}(\mathcal{D}_c), \quad (1)$$

with weights $(\lambda_G, \lambda_V, \lambda_D) = (0.3, 0.2, 0.5)$ that we fixed before running any experiment reported here. They were not tuned on the novel classes; Sec. 5 examines them on held-out validation classes. Ranks rather than raw values keep the heavy tails of $\mathcal{D}$ and $\mathcal{G}$ from deciding the sum.

The PMP schedule. We write S0 for supervised cross-entropy training of the ImageNet-initialised network on the base classes, R for the recovery epochs of the same loss (one on PlantVillage, two on the Earth-observation sets), M for episodic training, and P for a pruning step. Figure 1 shows the full schedule. After S0, P1 prunes to 60% of the parameters, R recovers, and M runs 1000 episodes (5-way, 5-shot, 15 queries) of prototypical training with frozen BatchNorm statistics, in the manner of Meta-Baseline [6]. While M runs we accumulate, for every mid channel, the mean norm of the episode-loss gradient with respect to its filter, G_{meta}. P2 prunes to the target budget with $\mathrm{DACIS}_c\,\big(1 + \gamma\, \mathrm{rk}(G_{\mathrm{meta},c})\big)$, γ=1, recomputed on the meta-trained network, and a last R repairs the damage. The alternatives we compare are P (prune once, then R), P→M, M→P and P→M→P→M. Two controls separate what a stage contributes. P→M with twice the episodes tests whether extra stages help only because they train longer. P→G→P replaces M by 200 episodes with the learning rate set to zero, so the network is untouched and only the gradient statistic G_{meta} is collected; it asks whether the information in the meta stage is useful without its weight updates. The learning rate of M was chosen on validation classes (Sec. 5).

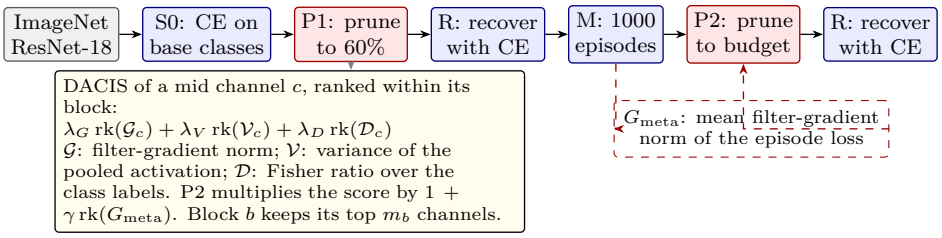


Figure 1. The PMP schedule and the DACIS score. Blue boxes are training stages (S0 and R use cross-entropy on the base classes, M is episodic), red boxes are pruning steps. P1 prunes to 60% of the parameters, M accumulates the mean filter-gradient norm G_{meta} of the episode loss, and P2 prunes to the target budget with a score that also uses G_{meta}.

4. Experimental protocol

Data. NWPU-RESISC45 [7] has 31,500 aerial RGB scenes of 256×256 pixels, 700 for each of 45 scene classes from airport and beach to wetland. We use its single-file Hugging Face copy and resize every scene to 128×128. EuroSAT-MS [18] has 27,000 Sentinel-2 [10] patches of 64×64 pixels (a 640 m footprint at 10 m resolution) with 13 spectral bands and 10 land-use classes. Each band is standardised with the mean and standard deviation of the base-class images. To accept 13 channels the first convolution repeats the ImageNet filters over the bands and scales them by 3/13; a control network sees only the visible bands B4, B3 and B2 through the unmodified filters. PlantVillage [20] supplies 54,305 colour photographs of single leaves on a plain background, 38 classes over 14 crops, resized to 128×128. PlantDoc [30] was assembled from internet photographs, so its leaves sit in cluttered scenes. We use the cropped-leaf classification release (2569 images, 28 label names); one label has two images and is dropped, which leaves 27 classes, and images are resized on the short side and centre-cropped to 128×128. No PlantDoc image is used for training or pruning.

Splits and episodes. A seed fixes a random partition of the classes. RESISC45 gives 25 base, 5 validation and 15 novel classes, EuroSAT-MS 5 base and 5 novel, PlantVillage 24 base, 6 validation and 8 novel. Seeds 0 to 4 give five different partitions per dataset, and every method sees the same one within a seed. PlantDoc is an evaluation set only: its 27 classes are used for 5-way episodes after training on PlantVillage base classes, so the test crosses a domain boundary. Because PlantDoc and PlantVillage share some concepts (tomato early blight, for one), this is a field-image test and not a strict novel-class test. Every table reports 5-way accuracy over 2000 episodes with 15 query images per class and $K \in \{1, 5, 10\}$ supports. The episodes are drawn once per dataset, seed and K and reused by all methods, which makes the comparisons paired.

Training and measurement. Stage S0 attaches a new linear head over the base classes and trains for 10 epochs (RESISC45), 8 (EuroSAT-MS) or 6 (PlantVillage) with SGD (Nesterov momentum 0.9, learning rate 0.02, weight decay $5{\cdot}10^{-4}$ on weight tensors, 3% linear warm-up then cosine decay, batch 128, label smoothing 0.1). Stage R reuses that head for 2 epochs at 0.01 on the two Earth-observation sets and one epoch on PlantVillage; the head is dropped at test time, where a parameter-free cosine prototype classifier is used. Stage M keeps BatchNorm statistics frozen, trains the cosine temperature (initialised to 10) and the other weights, and uses augmented batches. Stage M runs 1000 episodes (800 on EuroSAT-MS), two episodes per step, at a learning rate picked on the validation classes of the first PlantVillage partition from $\{10^{-4}, 2.5{\cdot}10^{-4}, 5{\cdot}10^{-4}, 10^{-3}, 2{\cdot}10^{-3}\}$ (it chose 0.0001, the smallest value of the grid); all datasets use that value. The same sweep on the RESISC45 validation classes was run as a check and not used for selection. Calibration for the scores uses 40 images per base class in batches of 64, with a fixed cosine temperature of 10. Training images get a random crop (scale 0.6 to 1) and random flips; PlantVillage and RESISC45 images also get brightness and contrast jitter. Mixed precision is on. Apart from that learning rate we tuned nothing, and nothing per method. Off-the-shelf baselines (MobileNetV3-Small, MobileNetV2 and EfficientNet-B0 with torchvision ImageNet weights) get the S0 recipe followed by R. All runs used one NVIDIA RTX PRO 6000 Blackwell Server Edition. Parameters and multiply-accumulates (MACs) are counted exactly on the pruned network; latency is measured in fp32 on that GPU (batch 128) and on a 20-thread server CPU (model not reported by the host) (one thread, batch 1, median of 60 runs after warm-up) and is not a statement about embedded hardware. With thirteen bands the dense trunk has 11.21 M parameters, and EuroSAT-MS budgets are relative to that. The compact networks use the torchvision feature trunk only.

Reporting. Tables give the mean over seeds of the per-seed mean accuracy, with the sample standard deviation over the 5 seeds as a subscript. This spread contains the effect of the class partition and of training noise. It is much larger than the sampling error of 2000 episodes (at most 0.2 points), so we quote it whenever a difference is small. For a pair of methods we give the mean paired difference in points with a 95% t-interval over the per-partition differences, and the number of partitions in which the sign agrees. Partitions, not episodes, are the replicates: episodes from one trained network are not independent draws of the training run.

5. Results

What 30% of the parameters costs. Partition decides the level. The dense network scores 86.9% on RESISC45 at 5 shots, 62.1% on EuroSAT-MS and 91.4% on PlantVillage, with single partitions on EuroSAT-MS ranging from 54.0% to 67.7% (Table 1). Keeping 30% of the parameters with DACIS under PMP costs 6.4 points on RESISC45 [+5.0,+7.8], 5.0 on EuroSAT-MS [+3.7,+6.4] and 4.9 on PlantVillage [+2.4,+7.3]; the brackets are 95% intervals over partitions. On PlantDoc field photographs the same PlantVillage networks fall from 55.4% to 47.5%.

The choice among scores is a small effect, and DACIS is not ahead. Under PMP the best and the worst of the eight scores are 2.7 points apart on RESISC45, 2.6 on EuroSAT-MS, 2.0 on PlantVillage and 3.0 on PlantDoc. Random selection ranks 3, 1, 2 and 6 of eight (1 is best). Of the 28 paired comparisons of a computed score with random selection (seven scores, four settings), 2 have a 95% interval above zero (Taylor on RESISC45; Taylor on PlantDoc) and 3 below it (BN scale on RESISC45; Fisher on RESISC45; BN scale on EuroSAT-MS); with that many intervals about 1.4 would exclude zero by chance alone; after a Bonferroni correction the smallest adjusted p value is 0.26 among the comparisons that favour a computed score and 0.01 among all of them, so the deficits of the worst scores (BatchNorm scale and Fisher ratio on RESISC45) are the only differences that survive it. The largest mean advantage is 1.9 points, and the intervals have a half-width of 1.7 points on average (up to 3.2), so five partitions cannot rule out differences of that size either way. Under PMP the first-order Taylor term is the only score ahead of random selection, on two settings: by +1.2 points on RESISC45 [+0.4,+2.0] and +1.9 on PlantDoc [+0.8,+3.0] under PMP, and by +2.5 on RESISC45 under P. DACIS minus random is −0.5 points on RESISC45 [−1.3,+0.2], −0.4 on EuroSAT-MS [−3.5,+2.7] and −0.3 on PlantVillage [−2.3,+1.8], and DACIS minus Taylor is −1.8 on RESISC45 [−2.3, −1.2] (Table 2). On the PlantVillage and EuroSAT-MS partitions the ranking of the scores hardly repeats (Kendall's W 0.24 and 0.22, against about 1/5 for unrelated rankings), while on RESISC45 it does (0.68) because Taylor is usually first and Fisher or BatchNorm scale last (Fig. 2). The channels the scores keep are far from identical (Jaccard overlap in the supplement), so the small differences do not come from the scores choosing the same channels.

Episodic training did not help the dense network, and the pruning schedules gain at most a point. Adding 1000 episodes (800 on EuroSAT-MS) of prototypical training after cross-entropy pre-training changes 5-shot accuracy of the dense network by −0.3 points on RESISC45 [−0.5,−0.1], −0.8 on EuroSAT-MS [−2.7,+1.0] and −0.3 on PlantVillage [−0.6,+0.1], at the learning rate chosen on validation classes (0.0001). In the validation sweep (supplement) 5-shot accuracy broadly falls as the learning rate grows, on both sets that have validation classes, and no value beats stopping after S0; our conclusion is for this recipe, with frozen BatchNorm and at most the smallest learning rate of the grid. Inside pruning schedules (Table 2) PMP is ahead of P by +0.7 points on RESISC45 [+0.2,+1.2] and level elsewhere. The gradient probe P→G→P, which collects G_{meta} without updating a weight in the meta stage, is ahead by +1.0 [+0.6,+1.3], as much as PMP. The gain therefore comes from the extra pruning step and recovery epoch that both schedules contain, not from episodic learning. Guo et al. [16] found that meta-learning methods trail simple fine-tuning on their crop-disease and EuroSAT targets.

Budget matters more than score, and a small pre-trained network beats all of them. Figure 3 shows accuracy falling steadily with the budget and the compact networks above and to the left of every pruned ResNet-18: on RESISC45 DACIS drops from 83.8% at 50% of the parameters to 71.9% at 10%, a larger step than the whole spread among scores. Pruning saves less time than operations (Table 3): at 30% of the parameters a ResNet-18 needs 0.251 G multiply-accumulates at 128×128 against 0.592 G for the dense network, which is 58% fewer, but its single-thread CPU latency is only 39% lower. We did not profile layer by layer. The stem, the shortcuts and the residual widths are untouched, and at batch size one the small layers may be limited by memory traffic and call overhead more than by arithmetic; either would explain the gap, and we cannot tell which. The off-the-shelf compact

	RESISC45		EuroSAT-MS		PlantVillage		PlantDoc	
Score at 30% of the parameters	P	PMP	P	PMP	P	PMP	P	PMP
Dense ResNet-18, S0 only	$86.9_{\pm 3.4}$		$62.1_{\pm 5.3}$		$91.4_{\pm 5.6}$		$55.4_{\pm 1.3}$	
Dense ResNet-18, S0+R+M+R	$86.7_{\pm 3.3}$		$61.7_{\pm 5.1}$		$91.3_{\pm 5.6}$		$55.5_{\pm 1.1}$	
Random	$79.7_{\pm 4.9}$	$81.0_{\pm 4.2}$	$55.6_{\pm 6.4}$	$57.6_{\pm 6.0}$	$85.6_{\pm 8.1}$	$86.8_{\pm 7.7}$	$46.6_{\pm 1.3}$	$46.7_{\pm 1.7}$
Filter ℓ_1	$80.5_{\pm 4.5}$	$81.0_{\pm 4.2}$	$56.2_{\pm 7.9}$	$57.0_{\pm 6.0}$	$85.8_{\pm 8.3}$	$86.4_{\pm 7.7}$	$48.7_{\pm 0.9}$	$48.1_{\pm 0.8}$
BN scale $\lvert\gamma\rvert$	–	$79.7_{\pm 4.2}$	–	$55.0_{\pm 6.1}$	$85.1_{\pm 8.9}$	$85.1_{\pm 8.7}$	$44.4_{\pm 1.7}$	$45.5_{\pm 1.0}$
Taylor	$82.2_{\pm 3.9}$	$82.2_{\pm 3.9}$	$58.1_{\pm 6.6}$	$57.0_{\pm 8.0}$	$88.3_{\pm 7.1}$	$87.1_{\pm 7.1}$	$47.9_{\pm 1.2}$	$48.5_{\pm 1.0}$
Gradient $\mathcal{G}$	–	$81.0_{\pm 4.5}$	–	$56.4_{\pm 6.5}$	$87.3_{\pm 7.2}$	$86.7_{\pm 7.9}$	$47.6_{\pm 1.5}$	$46.9_{\pm 2.5}$
Variance $\mathcal{V}$	–	$81.0_{\pm 4.0}$	–	$56.5_{\pm 5.2}$	$86.5_{\pm 6.9}$	$86.5_{\pm 7.3}$	$47.2_{\pm 1.4}$	$47.3_{\pm 1.3}$
Fisher $\mathcal{D}$	$79.3_{\pm 4.1}$	$79.5_{\pm 3.9}$	$55.8_{\pm 6.9}$	$56.0_{\pm 7.5}$	$85.2_{\pm 8.9}$	$85.3_{\pm 8.0}$	$46.8_{\pm 0.8}$	$46.4_{\pm 1.3}$
DACIS	$79.7_{\pm 4.1}$	$80.5_{\pm 4.2}$	$56.3_{\pm 6.7}$	$57.1_{\pm 4.9}$	$86.4_{\pm 7.7}$	$86.6_{\pm 7.5}$	$47.3_{\pm 1.4}$	$47.5_{\pm 1.6}$

Table 1. Eight channel scores at 30% of the parameters, 5-shot accuracy (%) on novel classes. P prunes once and recovers; PMP is the prune, meta-train, prune schedule of Fig. 1. Cells are the mean over 5 class partitions with the standard deviation over partitions as subscript. PlantDoc columns test the PlantVillage-trained networks on field photographs.

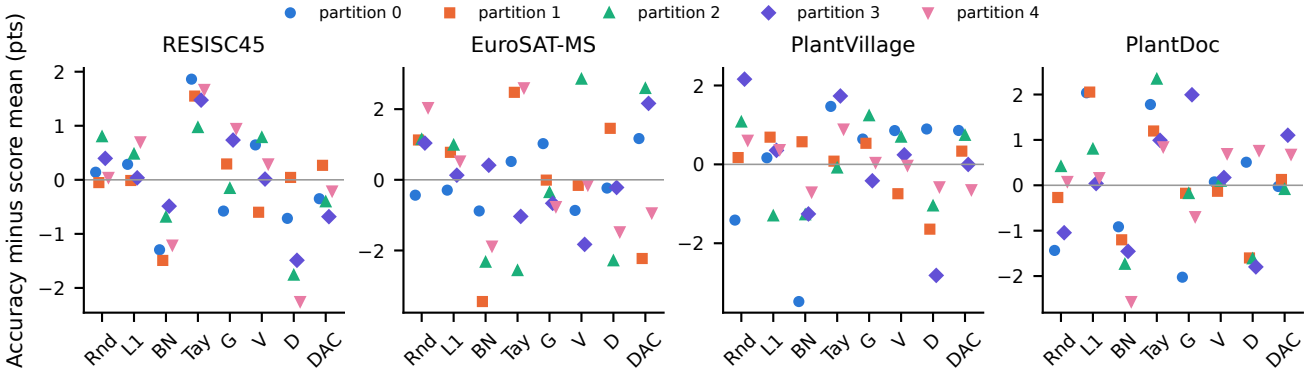


Figure 2. Accuracy of each score at 30% of the parameters (schedule PMP) on each class partition, relative to the mean over scores.

networks are a strong baseline that pruning studies in this setting should include. MobileNetV3-Small, with 0.93 M parameters, 0.018 G multiply-accumulates and 2.3 ms on the CPU, scores 88.1% on RESISC45, 72.4% on EuroSAT-MS and 91.1% on PlantVillage, which is +7.7 points above the 30% ResNet-18 on RESISC45 [+6.2,+9.1] and +15.3 on EuroSAT-MS [+9.4,+21.2], and above the unpruned ResNet-18 on EuroSAT-MS (62.1%). EfficientNet-B0 is higher still on RESISC45 and PlantVillage.

Sentinel-2 bands. On EuroSAT-MS the dense network with all thirteen bands scores 62.1% at 5 shots against 68.7% for the same network on the three visible bands (Table 4). In this naive adaptation, with five base classes and ImageNet filters repeated over the extra bands, the additional bands lowered novel-class accuracy; the interval over partitions is wide, and we do not conclude that spectral bands are unhelpful for few-shot EuroSAT. DACIS minus filter ℓ_1 under PMP is +0.1 points with thirteen bands and −0.3 with three.

Recovery and the parts of DACIS (PlantVillage). A longer recovery after pruning did little: with six epochs instead of one, random selection and filter ℓ_1 reach 86.3% and 87.3% under schedule P on PlantVillage, against 85.6% and 85.8% with one. Of the 9 variants of DACIS in the supplement (single components, pairs, no meta-gradient term, other depth profile or scaling), the largest mean change in PlantVillage PMP accuracy is 1.3 points, below the full score, for the Fisher ratio alone; 1 of them have a 95% interval clear of zero. On the validation classes, the eight weightings of Eq. (1) that we tried span 87.9% to 88.9% at 5 shots against 88.5% for the weights fixed in advance, so we left λ alone; this is a sensitivity check on validation classes, not a tuning protocol.

Difference (a − b), points	RESISC45	EuroSAT-MS	PlantVillage	PlantDoc
Scores, schedule P				
DACIS − Random	+0.1 [−1.4,+1.5]	+0.7 [−1.6,+2.9]	+0.8 [−1.3,+3.0]	+0.7 [−1.2,+2.6]
DACIS − Filter ℓ_1	−0.8 [−2.0,+0.4]	+0.1 [−3.6,+3.9]	+0.6 [−1.9,+3.1]	−1.4 [−3.3,+0.5]
DACIS − Taylor	−2.4 [−3.1,−1.7]	−1.8 [−5.2,+1.7]	−1.9 [−2.9,−0.9]	−0.6 [−1.6,+0.4]
Scores, schedule PMP				
DACIS − Random	−0.5 [−1.3,+0.2]	−0.4 [−3.5,+2.7]	−0.3 [−2.3,+1.8]	+0.8 [−0.4,+2.1]
DACIS − Filter ℓ_1	−0.6 [−1.2,+0.0]	+0.1 [−2.6,+2.9]	+0.2 [−1.3,+1.7]	−0.7 [−2.4,+1.1]
DACIS − Taylor	−1.8 [−2.3,−1.2]	+0.2 [−5.1,+5.4]	−0.6 [−1.9,+0.8]	−1.1 [−2.4,+0.3]
Schedules, DACIS at 30%				
PMP − P	+0.7 [+0.2,+1.2]	+0.8 [−1.9,+3.6]	+0.1 [−0.7,+0.9]	+0.1 [−0.8,+1.1]
P→M − P	−0.0 [−0.9,+0.8]	−0.5 [−2.7,+1.8]	−0.7 [−2.4,+1.0]	−0.8 [−1.6,−0.0]
M→P − P	+0.7 [−0.1,+1.6]	+0.2 [−1.2,+1.6]	+0.3 [−1.2,+1.7]	−0.7 [−1.5,+0.2]
P→G→P (no weight update) − P	+1.0 [+0.6,+1.3]	+0.3 [−1.5,+2.1]	+0.3 [−0.4,+1.0]	−0.2 [−0.8,+0.5]
P→M→P→M − PMP	–	–	−0.1 [−1.2,+1.1]	+0.1 [−1.0,+1.2]
P→M (2× episodes) − P→M	–	–	−0.1 [−0.7,+0.4]	−0.0 [−1.0,+0.9]
Episodic stage on the dense network				
Dense+M − Dense	−0.3 [−0.5,−0.1]	−0.8 [−2.7,+1.0]	−0.3 [−0.6,+0.1]	−0.0 [−0.4,+0.3]

Table 2. Paired differences in points (mean over partitions, with the 95% t-interval over partitions in brackets); (a − b) is positive when a is more accurate. Test episodes are shared inside a partition. A dash marks a comparison that was not run on that dataset.

Model	Params (M)	MACs (G)	CPU ms	GPU img/s	RESISC45	EuroSAT-MS	PlantVillage
ResNet-18, dense	11.18	0.592	10.8	36520	$86.9_{\pm 3.4}$	$62.1_{\pm 5.3}$	$91.4_{\pm 5.6}$
ResNet-18, 50% (DACIS)	5.59	0.398	8.6	38034	$83.8_{\pm 3.7}$	$58.6_{\pm 5.7}$	$88.6_{\pm 7.2}$
ResNet-18, 30% (DACIS)	3.35	0.251	6.6	40431	$80.5_{\pm 4.2}$	$57.1_{\pm 4.9}$	$86.6_{\pm 7.5}$
ResNet-18, 20% (DACIS)	2.24	0.178	5.7	41645	$77.4_{\pm 4.9}$	$54.5_{\pm 8.1}$	$83.8_{\pm 8.2}$
ResNet-18, 10% (DACIS)	1.11	0.105	4.6	45865	$71.9_{\pm 4.8}$	$51.4_{\pm 7.5}$	$78.2_{\pm 10.2}$
MobileNetV3-Small, dense	0.93	0.018	2.3	63688	$88.1_{\pm 3.3}$	$72.4_{\pm 4.9}$	$91.1_{\pm 5.9}$
MobileNetV2, dense	2.22	0.098	4.3	24416	$86.9_{\pm 3.1}$	$65.4_{\pm 3.5}$	$91.2_{\pm 6.3}$
EfficientNet-B0, dense	4.01	0.126	6.3	18928	$90.5_{\pm 2.4}$	$69.9_{\pm 5.1}$	$93.8_{\pm 5.2}$

Table 3. Cost and 5-shot accuracy (%). The pruned architecture is the same for every score, so one cost row serves all scores; accuracy is for DACIS under PMP. Cost is for a 128×128 RGB input: CPU is single-thread latency at batch size 1 on a 20-thread server CPU (model not reported by the host), GPU throughput is at batch size 128 on NVIDIA RTX PRO 6000 Blackwell Server Edition. Latency is measured, not a statement about embedded devices. The accuracy columns use each dataset's own input size and bands. The compact networks are trained with the same recipe and are not pruned.

6. Limitations

5 class partitions are few. The spread over partitions in the tables mixes how hard a partition is with training noise, and it is itself estimated from five numbers. Every hyper-parameter, including λ, was fixed in advance and shared by all scores. The one value we tuned, the episodic learning rate, was tuned on the dense network and on PlantVillage validation classes, then reused on the Earth-observation sets; a score or a stage that needs its own learning rate or recovery schedule did not get its best case.

"Novel" means novel with respect to the base classes, not with respect to pre-training: the ImageNet initialisation has seen photographs of airplanes, ships, tennis courts and other concepts that reappear among the novel RESISC45 classes. The Earth-observation results cover two classification benchmarks of single-label patches. EuroSAT-MS has ten classes, which allows a 5/5 class split and nothing larger, so its novel-class numbers rest on a five-class base set. RESISC45 is RGB. No model uses coordinates, acquisition dates, synthetic-aperture radar or any sensor beyond the thirteen optical bands, and none was tested on full scenes or on time series, so what is specific to geospatial data is absent from this study. PlantVillage images are leaves on a plain background, and the dataset holds several photographs of the same leaf. We did not use leaf identities to keep support and query images apart,

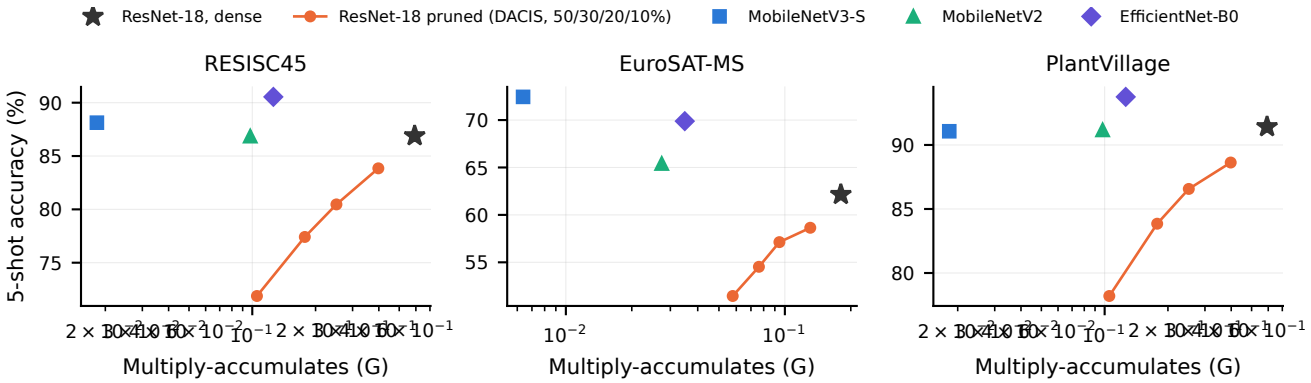


Figure 3. 5-shot accuracy against multiply-accumulates (log scale), mean over 5 partitions. The orange line connects the ResNet-18 pruned with DACIS to 50, 30, 20 and 10% of the parameters; the compact networks are not pruned. Accuracy against budget for four scores is in the supplement.

EuroSAT-MS, novel classes	1-shot	5-shot	10-shot
13 bands, dense (S0)	$49.4_{\pm 5.5}$	$62.1_{\pm 5.3}$	$64.9_{\pm 5.0}$
RGB bands, dense (S0)	$53.7_{\pm 4.3}$	$68.7_{\pm 3.9}$	$71.4_{\pm 3.6}$
13 bands, dense (S0+R+M+R)	$49.0_{\pm 5.2}$	$61.7_{\pm 5.1}$	$64.3_{\pm 5.0}$
RGB bands, dense (S0+R+M+R)	$53.1_{\pm 4.5}$	$68.1_{\pm 3.8}$	$70.9_{\pm 3.4}$
13 bands, filter ℓ_1, PMP 30%	$46.1_{\pm 6.5}$	$57.0_{\pm 6.0}$	$59.3_{\pm 5.2}$
RGB bands, filter ℓ_1, PMP 30%	$49.8_{\pm 2.4}$	$63.9_{\pm 2.6}$	$66.6_{\pm 2.3}$
13 bands, DACIS, PMP 30%	$46.0_{\pm 5.4}$	$57.1_{\pm 4.9}$	$59.4_{\pm 4.1}$
RGB bands, DACIS, PMP 30%	$50.2_{\pm 3.5}$	$63.6_{\pm 3.9}$	$66.2_{\pm 3.7}$
13 bands, MobileNetV3-Small	$57.7_{\pm 7.7}$	$72.4_{\pm 4.9}$	$75.7_{\pm 4.5}$
RGB bands, MobileNetV3-Small	$58.3_{\pm 3.7}$	$74.7_{\pm 4.1}$	$78.0_{\pm 4.1}$
13 bands − RGB, dense (5-shot)	−6.6 [−17.4,+4.2]		
13 bands − RGB, DACIS PMP (5-shot)	−6.4 [−15.5,+2.6]		

Table 4. EuroSAT-MS, novel classes: thirteen bands against the three visible bands. The extra bands lower accuracy in this adaptation.

so near-duplicates can raise the PlantVillage numbers; the PlantDoc, RESISC45 and EuroSAT-MS results do not depend on that. PlantDoc here is a cropped-leaf release with one dropped class, and its concepts overlap those of PlantVillage.

Cost is reported as parameters, multiply-accumulates, and latency on one server CPU and one GPU. We did not run on a Raspberry Pi, a Jetson or an accelerator on board a satellite, and we measured no energy, so we say nothing about those devices. Only ResNet-18 and only the inner convolution of each residual block were pruned. Quantisation, distillation, architecture search and recent few-shot learners were not compared; the claims are limited to channel scores and training schedules inside this one design.

What we can and cannot conclude. The paired intervals over five partitions are wide, so a null result means "not distinguishable at about 1.7 points", not "equal". Each partition has one training run per method, so training noise and method effect are confounded inside a partition, and the 28 comparisons share their random baseline and their partitions and are not independent. The episodic stage was tested with one recipe (frozen BatchNorm, at most the smallest learning rate of the grid, 1000 episodes, 5-way 5-shot). The compact networks were trained with a recipe set for the ResNet and were not pruned, so the comparison with the pruned ResNet-18 is not at equal effort, and we did not try satellite-pretrained backbones. RESISC45 scenes were resized from 256 to 128 pixels, which changes the ground sampling distance, and the class-disjoint split does not separate neighbouring tiles geographically. The Sentinel-2 comparison between thirteen and three bands depends on how the first convolution was initialised.

7. Conclusion

For a ResNet-18 that has to recognise unseen classes from a few examples, which channels were removed mattered little. Under PMP at 30% of the parameters eight scores, random selection among them, land 2.0 to 3.0 points apart on four test settings. DACIS cannot be told from random selection, and the earlier preprint's claim for it does not carry over to held-out classes. The Taylor term was ahead on two settings, by at most 1.9 points, a difference that does not survive correction for the number of comparisons. The meta stage did not raise the accuracy of the dense network, and the gain of two-stage schedules is matched by a probe that updates no weight. The numbers moved most with the class partition, the budget and the architecture: an unpruned MobileNetV3-Small beat the pruned ResNet-18 on both Earth-observation sets, and pruning bought less time (39% lower CPU latency) than the operation count promised (58% fewer multiply-accumulates).

For few-shot classification of optical patches like those of RESISC45 and EuroSAT, we would

start from a small pretrained architecture, and if a ResNet-18 must be pruned, use random selection or the Taylor term, skip the meta stage, and report results over several class-disjoint partitions with paired episodes. We did not prune the compact networks, try Earth-observation pretrained backbones, or measure anything on embedded hardware, so those questions remain open, as does whether a score computed on the target classes would do better.

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